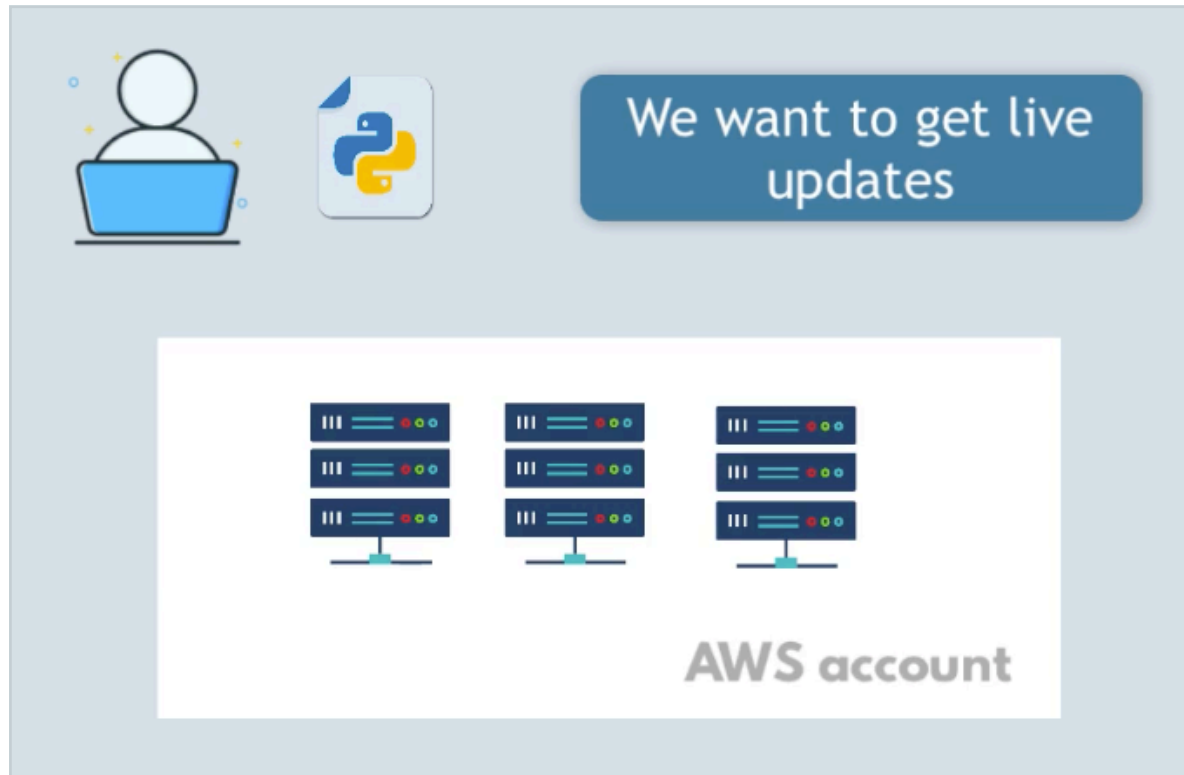


Module 14 - chapter 6

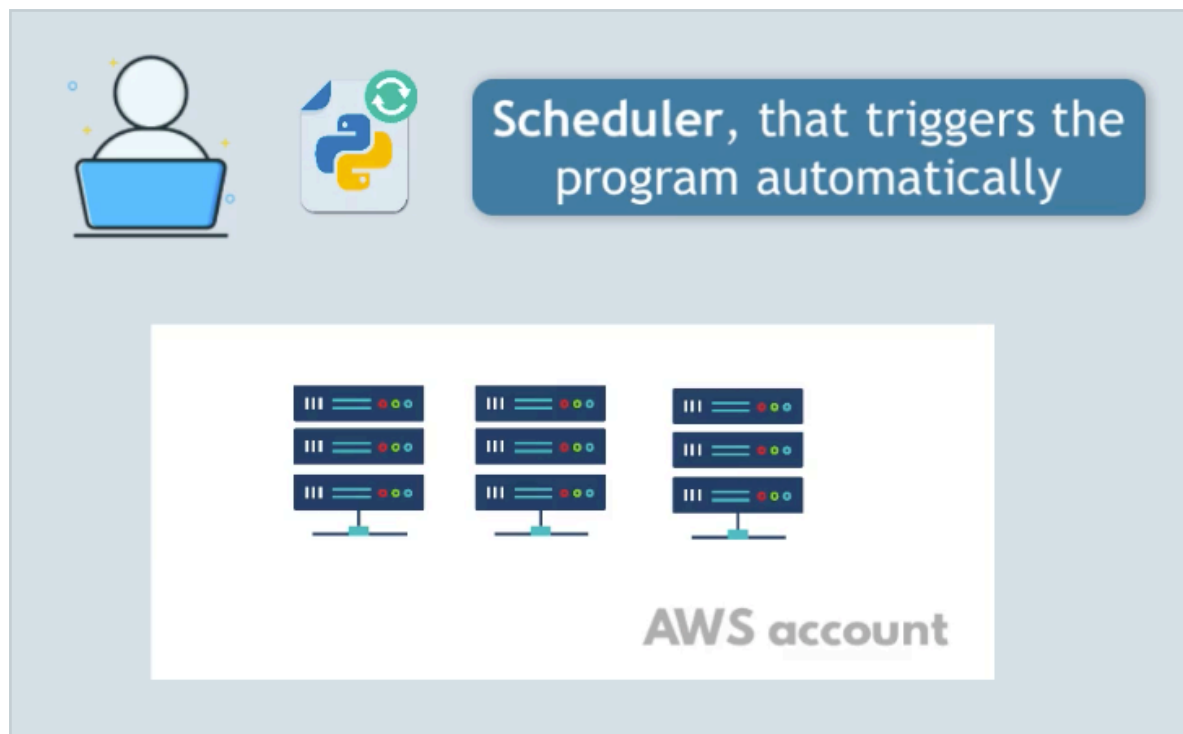
Write a Scheduled Task in Python

So the status of the EC2 instances will change so executing the program once to check the status doesn't make sense, we want to monitor and get live updates of the instance statuses. So executing the program manually also doesn't make sense.



Scheduler

So we want a scheduler -> that triggers the program automatically, for example to run the program every 10 minutes or every hour.



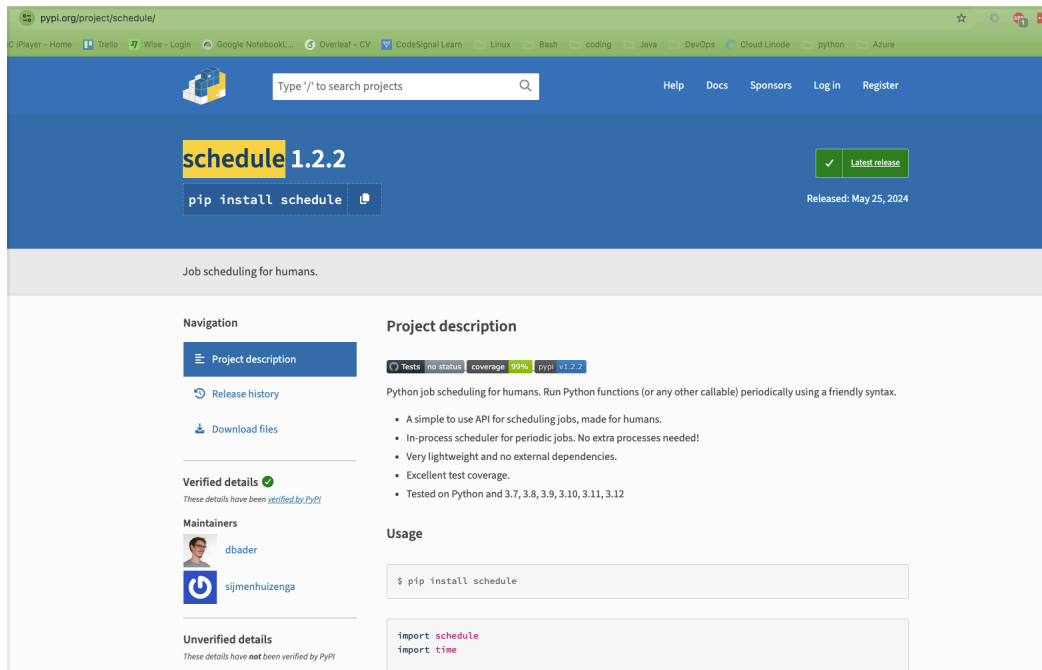
And with python programs we can use them as scheduled tasks.

We will write a scheduler to run our python program every 5 minutes

Implementation

First we add a scheduler library to our project -> schedule

-> `pip install schedule`



```
Terminal  zsh (/usr/local/bin/zsh)  x  +  v

> pip install schedule
Collecting schedule
  Downloading schedule-1.2.2-py3-none-any.whl.metadata (3.8 kB)
  Downloading schedule-1.2.2-py3-none-any.whl (12 kB)
Installing collected packages: schedule
Successfully installed schedule-1.2.2
```

And I import the library into our code

```
main.py x
1  import boto3
2  import schedule  You, Moments ago • Uncommitted changes
3
4  ec2_client = boto3.client(*args: 'ec2', region_name='eu-west-3')
5  ec2_resource = boto3.resource(*args: 'ec2', region_name='eu-west-3')
6
7  def check_instance_status(): new *
8      statuses = ec2_client.describe_instance_status()
9      for status in statuses['InstanceStatuses']:
10         ins_status = status['InstanceStatus']['Status']
11         sys_status = status['SystemStatus']['Status']
12         state = status['InstanceState']['Name']
13         print(f"Instance {status['InstanceId']} is {state} with instance status {ins_status} and system
```

(Also refactored the status check to be inside a function 🙌)

Now we create the scheduler object:

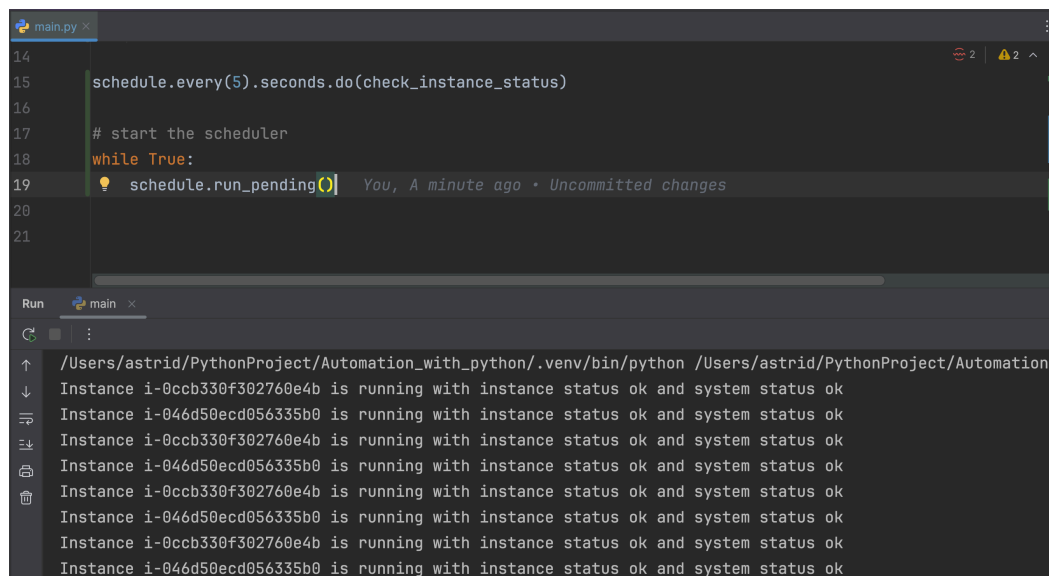
-> `schedule.every(5).minutes.do(check_instance_status)`

```

7     def check_instance_status(): 1 usage new *
8         statuses = ec2_client.describe_instance_status()
9         for status in statuses['InstanceStatuses']:
10             ins_status = status['InstanceState']['Status']
11             sys_status = status['SystemStatus']['Status']
12             state = status['InstanceState']['Name']
13             print(f"Instance {status['InstanceId']} is {state} with ins
14
15     schedule.every(5).minutes.do(check_instance_status)
16
17     # start the scheduler
18     while True:
19         schedule.run_pending()

```

To test the demo we will use seconds so we can see the scheduler working sooner:



The screenshot shows a code editor with a file named 'main.py'. The code in the editor is as follows:

```

14
15     schedule.every(5).seconds.do(check_instance_status)
16
17     # start the scheduler
18     while True:
19         schedule.run_pending()
20
21

```

Below the editor is a terminal window titled 'Run' showing the output of the script. The output consists of multiple lines, each reporting the status of an EC2 instance. The instances are identified by their IDs: i-0ccb330f302760e4b and i-046d50ecd056335b0. The output for each instance is: 'Instance i-0ccb330f302760e4b is running with instance status ok and system status ok' and 'Instance i-046d50ecd056335b0 is running with instance status ok and system status ok'. The terminal shows that the script is running successfully and checking the status of the instances every 5 seconds.

But seconds is too much, we will end up overloading the system.

So we will change the code back to minutes:

```

15     schedule.every(5).minutes.do(check_instance_status)
16
17     # start the scheduler
18     while True:
19         schedule.run_pending()

```

We can also have every hour
-> schedule.every().hour

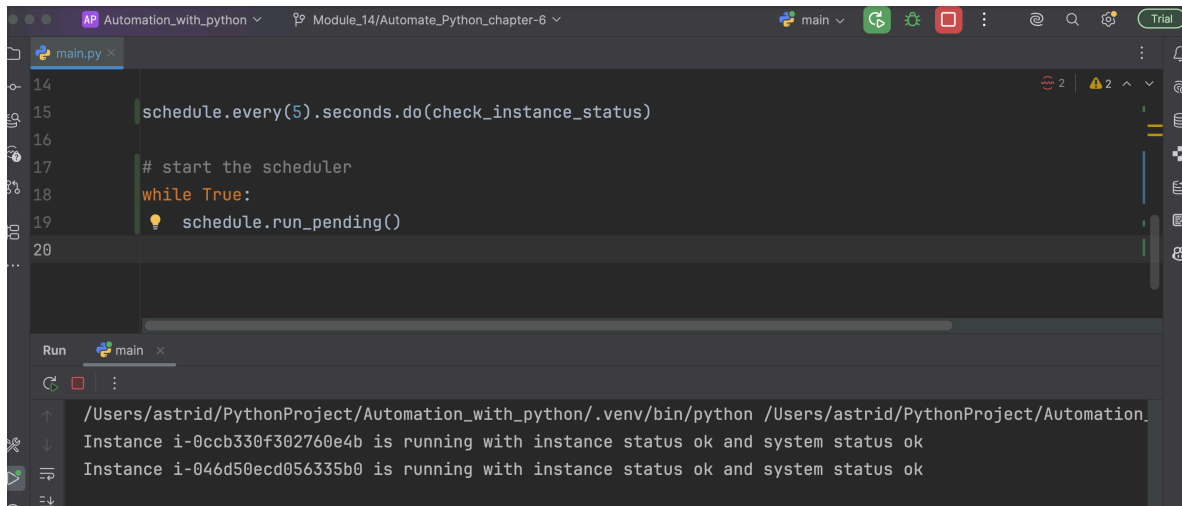
Or every day at specific time

-> `schedule.every().day.at("1:00")`

Or every Monday

-> `schedule.every().monday.at("12:00")`

So now let's run the scheduler program using seconds



The screenshot shows a code editor with a file named `main.py`. The code defines a function `check_instance_status` (lines 14-16) and a `while True` loop (lines 17-19) that calls `schedule.run_pending()` every 5 seconds. Below the code editor, the 'Run' console shows the output of the script, indicating that two EC2 instances are running with 'ok' status.

```
14 def check_instance_status():
15     # ... (code omitted) ...
16
17 # start the scheduler
18 while True:
19     schedule.run_pending()
20
```

Run main x

```
/Users/astrid/PythonProject/Automation_with_python/.venv/bin/python /Users/astrid/PythonProject/Automation.
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
```

and then add another EC2 instance to the infra

```

120 resource "aws_instance" "myapp-server" {...}
139
140 resource "aws_instance" "myapp-server-two" {...}
159
160 resource "aws_instance" "myapp-server-three" {
161     ami = data.aws_ssm_parameter.latest-amazon-linux-image.value
162     instance_type = var.instance_type
163
164     subnet_id = aws_subnet.myapp-subnet-1.id
165     vpc_security_group_ids = [aws_security_group.myapp-sg.id]
166     availability_zone = var.avail_zone
167
168     associate_public_ip_address = true
169     key_name = aws_key_pair.ssh-key.key_name
170
171     tags = {
172         Name = "${var.env_prefix}-server-three"
173     }
174
175     user_data = file("entry-script.sh")
176

```

-> Terraform plan

```

Terminal: Local x + v
+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform c
apply" now.

```

-> terraform apply

```

}

Plan: 1 to add, 0 to change, 0 to destroy.
aws_instance.myapp-server-three: Creating...
aws_instance.myapp-server-three: Still creating... [00m10s elapsed]
aws_instance.myapp-server-three: Creation complete after 13s [id=i-09852f1af77c49e7b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

server-ip = "13.36.167.207"
~/terraform ▶ Module_14/Au...on_chapter-5 ?1 ▶

```

and then look at the logs:

```

Run main x
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status initializing
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status initializing
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status initializing
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok

```

```

Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status initializing
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status initializing
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok

```

And now the last server is ready:

```

Instance i-09852f1af77c49e7b is running with instance status initializing and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status initializing and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok

```

Next, let's remove two servers.

First notice that `.describe_instance_status()` show running servers and if we remove servers we won't get those in the logs.

We can include servers that are terminated as well by overriding the default parameter that shows only running servers and for that we use
 -> **IncludeAllInstances** parameter

docs.aws.amazon.com/boto3/latest/reference/services/ec2/client/describe_instance_status.html

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aws

Boto3 1.43.40 documentation

Q Search

- VPCLattice
- WAF
- WAFRRegional
- WAFV2
- WellArchitected
- WickrAdminAPI
- ConnectWisdomService
- WorkDocs
- WorkMail
- WorkMailMessageFlow
- WorkSpaces

A filter name and value pair that is used to return a more specific list of results from a describe operation. Filters can be used to match a set of resources by specific criteria, such as tags, attributes, or IDs.

If you specify multiple filters, the filters are joined with an **AND**, and the request returns only results that match all of the specified filters.

For more information, see [List and filter using the CLI and API](#) in the *Amazon EC2 User Guide*.

- **Name (string)** –
The name of the filter. Filter names are case-sensitive.
- **Values (list)** –
The filter values. Filter values are case-sensitive. If you specify multiple values for a filter, the values are joined with an **OR**, and the request returns all results that match any of the specified values.
 - (string) –
- **IncludeAllInstances (boolean)** –
When **true**, includes the health status for all instances. When **false**, includes the health status for running instances only.
Default: **false**

RETURN TYPE:
dict

RETURNS:
Response Syntax

```

-> ec2_client.describe_instance_status(
    IncludeAllInstances=True
)

```

```

main.py x
7     def check_instance_status(): 1 usage new *
8     statuses = ec2_client.describe_instance_status( You, 19 minutes ago • Un
9         IncludeAllInstances=True
10    )
11    for status in statuses['InstanceStatuses']:
12        ins_status = status['InstanceStatus']['Status']

```

This is what my AWS currently has:

Instances (3) Info Last updated less than a minute ago Connect Instance state Actions Launch

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	dev-server	i-0ccb330f302760e4b	Running	t2.micro	2/2 checks passed	eu-west-3a	-	13.36.167.207	-
☐	dev-server-three	i-09852f1af77c49e7b	Running	t2.micro	2/2 checks passed	eu-west-3a	-	51.44.167.234	-
☐	dev-server-two	i-046d50ecd056335b0	Running	t2.micro	2/2 checks passed	eu-west-3a	-	13.38.27.206	-

So I run the scheduler

```

Run main x
/Users/astrid/PythonProject/Automation_with_python/.venv/bin/python /Users/astrid/PythonProject
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
#####
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
#####

```

then I remove 2 servers using terraform

```

115
116     output "server-ip" {
117         value = aws_instance.myapp-server.public_ip
118     }
119
120     resource "aws_instance" "myapp-server" {...}
139
140
141     Pardo, Today • [Module_14] Edit TF for chapter 5

```

-> terraform plan

Plan: 0 to add, 0 to change, 2 to destroy.

-> terraform apply

And they start to shut down:

```
Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is running with instance status ok and system status ok
Instance i-046d50ecd056335b0 is running with instance status ok and system status ok
#####

Instance i-0ccb330f302760e4b is running with instance status ok and system status ok
Instance i-09852f1af77c49e7b is shutting-down with instance status not-applicable and system status not-ap
Instance i-046d50ecd056335b0 is shutting-down with instance status not-applicable and system status not-ap
#####
```

```
0ccb330f302760e4b is running with instance status ok and system status ok
09852f1af77c49e7b is shutting-down with instance status not-applicable and system status not-applicable
046d50ecd056335b0 is shutting-down with instance status not-applicable and system status not-applicable
#####

0ccb330f302760e4b is running with instance status ok and system status ok
09852f1af77c49e7b is shutting-down with instance status not-applicable and system status not-applicable
046d50ecd056335b0 is shutting-down with instance status not-applicable and system status not-applicable
#####
```

Instances (3) Info							Last t less than a min
Saved filter sets Choose filter set ▼		Find Instance by attribute or tag (case-sensitive)					
☐	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Z	
☐	dev-server	i-0ccb330f302760e4b	✓ Running	t2.micro	✓ 2/2 checks passec	eu-west-3a	
☐	dev-server-three	i-09852f1af77c49e7b	⌚ Shutting-d...	t2.micro	-	eu-west-3a	
☐	dev-server-two	i-046d50ecd056335b0	⌚ Shutting-d...	t2.micro	-	eu-west-3a	

And now the servers are terminated:

```
0ccb330f302760e4b is running with instance status ok and system status ok
09852f1af77c49e7b is terminated with instance status not-applicable and system status not-applicable
046d50ecd056335b0 is terminated with instance status not-applicable and system status not-applicable
#####
```

Instances (3)

Info

Last updated less than a minute ago

Connect

Instance state

Actions

Saved filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elas
☐	dev-server	I-0ccb330f302760e4b	Running	t2.micro	2/2 checks passed	eu-west-3a	-	13.36.167.207	-
☐	dev-server-three	I-09852f1af77c49e7b	Terminated	t2.micro	-	eu-west-3a	-	-	-
☐	dev-server-two	I-046d50ecd056335b0	Terminated	t2.micro	-	eu-west-3a	-	-	-